

CHAPTER 04

LAND AND CLIMATE OF PAKISTAN



What do you understand by climate?

The long-lasting conditions of weather in a particular area such as air pressure, temperature, humidity and rainfall are termed as the climate of an area. The climate of Pakistan is dry and hot. Climate refers to the long-term patterns and averages of weather conditions in a particular region, including temperature, precipitation, humidity, wind patterns, and other atmospheric variables. It's distinct from weather, which describes short-term atmospheric conditions. Climate is determined by factors such as latitude, elevation, proximity to bodies of water, ocean currents, and topography, as well as human influences like greenhouse gas emissions. Climate change refers to significant and lasting alterations in these long-term patterns, often attributed to human activities such as burning fossil fuels and deforestation, which release greenhouse gases into the atmosphere and contribute to global warming.

Pakistan's climate varies significantly across its diverse geographical regions, influenced by factors such as altitude, proximity to bodies of water, and topography. Generally, Pakistan experiences four distinct seasons:

1. Winter (December to February): In the northern regions, winters are cold with heavy snowfall in mountainous areas like Gilgit-Baltistan and Khyber Pakhtunkhwa. In the plains, temperatures are cooler, particularly at night, but snowfall is rare except in some high-altitude areas.

2. Spring (March to May): Spring brings mild to warm temperatures across most of Pakistan. This season is characterized by blooming flowers, particularly in the northern regions, and comfortable weather for outdoor activities.

3. Summer (June to August): Summers are hot and dry, especially in the plains and southern regions like Sindh and Balochistan. Temperatures often soar above 40°C (104°F) in these areas, while the northern regions remain relatively cooler due to their higher elevation.

4. Autumn (September to November): Autumn is a transitional season with gradually decreasing temperatures. It's generally pleasant, with clear skies and cooler weather, making it a popular time for tourism, particularly in the northern areas.

The climate also varies between the eastern and western parts of the country. The western regions, including Balochistan, experience arid to semi-arid conditions with minimal rainfall, while the eastern parts, including Punjab and Sindh, have a more humid climate with higher rainfall, particularly during the monsoon season (July to September).

However, Pakistan's climate is also affected by broader global patterns, including monsoon activity originating from the Indian Ocean and climate change impacts such as rising temperatures and changing precipitation patterns. These factors can contribute to variability and unpredictability in Pakistan's weather and climate.

DIFFERENT CLIMATE REGIONS OF PAKISTAN

SUB-TROPICAL CONTINENTAL HIGHLAND

It includes Pakistan's north eastern and northwestern mountain ranges. Here the winter is extremely cold. Summer season is cool. In spring and winter season, it is mostly foggy and rainy.

SUB-TROPICAL CONTINENTAL HIGHLAND

In this climatic region most of the parts of Baluchistan are included from May to the mid of September hot and dust winds continuously blow. The hottest areas of Pakistan Sibbi and Jacobabad are located in this region.

SUB-TROPICAL CONTINENTAL HIGHLAND

It consists of Punjab and Sindh. The summer season is very hot here. The north of Punjab receives late summer monsoon rains while the rest of Punjab receives less fall. Thal, Kachhi, Sibbi and south eastern plains are dry and a large area in desert.

SUB-TROPICAL CONTINENTAL HIGHLAND

This climatic region includes the coastal areas of Sindh and Baluchistan provinces, or the southern part of Pakistan. The temperature is moderate. Annual average temperature is 32°C and rainfall is 180 mm. May and June are the hottest months

SUB-TROPICAL CONTINENTAL HIGHLAND

The climate has a profound impact on human life in various ways, influencing everything from health and agriculture to economy and culture. Here are some of the significant effects of climate on human life:

1. Health Impacts:

- Extreme temperatures can lead to heat-related illnesses such as heatstroke and dehydration during heatwaves, or hypothermia during cold spells.
- Changes in climate patterns can affect the spread of vector-borne diseases like malaria and dengue fever, as well as waterborne illnesses.
- Poor air quality, exacerbated by climate factors such as temperature inversions and pollution, can lead to respiratory problems and cardiovascular diseases.

2. Agriculture and Food Security:

- Climate conditions like temperature, rainfall, and humidity directly affect crop yields and livestock productivity.
- Shifts in climate patterns, such as changes in rainfall timing or intensity, can lead to crop failures, affecting food availability and prices.
- Extreme weather events like droughts, floods, and storms can devastate agricultural systems, leading to food shortages and economic losses.

3. Water Resources:

- Changes in precipitation patterns and temperatures impact water availability for drinking, irrigation, and industrial purposes.
- Melting glaciers and changes in snowmelt patterns affect river flows, potentially leading to water scarcity in regions dependent on glacial meltwater.

4. Economic Effects:

- Climate-related disasters, such as hurricanes, floods, and wildfires, can cause extensive damage to infrastructure, homes, and businesses, leading to economic losses and disruptions.
- Industries like tourism, agriculture, and fishing are particularly vulnerable to climate variability and extreme weather events.

5. Migration and Displacement:

- Climate change can exacerbate environmental pressures, leading to displacement of populations due to factors like sea-level rise, desertification, and loss of arable land.
- Climate-induced migration can strain resources and infrastructure in receiving areas, leading to social tensions and conflicts.

6. Cultural and Social Impacts:

- Indigenous communities and cultures closely tied to the land may face disruptions due to changing climate conditions, affecting traditional practices and knowledge.
- Climate-related disasters can lead to displacement of communities, loss of cultural heritage, and disruption of social networks.

Overall, the impact of climate on human life is multifaceted and complex, affecting various aspects of society, economy, and environment. Addressing climate change and adapting to its impacts are critical for ensuring the well-being and resilience of communities around the world.

ENVIRONMENTAL POLLUTION:

It refers to the introduction of harmful substances or contaminants into the natural environment, causing adverse effects on ecosystems, human health, and biodiversity. Pollution can take many forms and originate from various sources, both natural and human-made. Some common types of environmental pollution include:

1. AIR POLLUTION

- Includes pollutants such as particulate matter (PM), nitrogen oxides (NO_x), sulfur dioxide (SO₂), carbon monoxide (CO), volatile organic compounds (VOCs), and ozone (O₃).
- Leads to respiratory diseases, cardiovascular problems, and contributes to climate change.

2. WATER POLLUTION

- Resulting from discharge of untreated sewage, industrial effluents, agricultural runoff, and improper waste disposal.
- Contaminants include heavy metals, pathogens, pesticides, fertilizers, and plastic waste.
- Causes waterborne diseases, ecosystem degradation, and threatens aquatic biodiversity.

3. SOIL POLLUTION

- Occurs due to contamination by industrial waste, mining activities, use of pesticides and fertilizers, and improper disposal of hazardous materials.
- Includes pollutants such as heavy metals, chemicals, and organic pollutants.
- Impairs soil fertility, affects plant growth, and can lead to groundwater contamination.

4. NOISE POLLUTION

- Caused by excessive noise from transportation, industrial activities, construction, and urbanization.
- Adversely affects human health, leading to hearing loss, stress, sleep disturbances, and communication problems.
- Disrupts wildlife behavior and habitats.

5. LIGHT POLLUTION

- Arises from excessive or misdirected artificial light from urban areas, outdoor lighting, and advertising.
- Disrupts natural light-dark cycles, affecting ecosystems, wildlife behavior, and human health, including sleep patterns and circadian rhythms.

6. PLASTIC POLLUTION

- Involves accumulation of plastic waste in the environment, particularly in oceans, rivers, and terrestrial ecosystems.
- Harms marine life through ingestion, entanglement, and habitat destruction.
- Contributes to microplastic contamination in food chains and poses risks to human health.

Addressing environmental pollution requires coordinated efforts at local, national, and global levels, including implementing regulations, adopting cleaner technologies, promoting sustainable practices, and raising awareness about environmental conservation.

PHYSICAL FEATURES OF PAKISTAN

The land of Pakistan is divided into the following four physical features:

1) MOUNTAIN

Any part of land which is 3,000 feet high and whose sides are steep is called a mountain. The mountain of Pakistan is divided into two parts:

- Northern and North Eastern Mountains: This include (i) Himalayas (ii) Karakoram
- North-Western and Western Mountains: Western mountains are less in height than northern mountains. These ranges are divided into following parts: a) The Hindukash Mountains b) The Koh-i-Safed Range c) The Waziristan Hills d) The Suleman Hills

2) PLATEAUS

The following two plateaus are located in Pakistan:

- The Potwar Plateau: This plateau lies between the river Indus and Jhelum.
- Balochistan Plateau: It is located in the west of Kirthar and Suleman range.

3) PLAINS

The plains are divided into three parts.

- The upper Indus plain: It is the part of Indus plain above Mithan Kot.
- The lower Indus plain: It is the area of south of Mithan Kot.
- The Indus delta: The River Indus forms its delta near Thatta.

4) THE DESERTS INCLUDING COASTAL AREAS

- Thal: It includes Mianwali, Muzaffargarh, and Dera Ghazi Khan. Huge sand dunes are found here.
- Cholistan: 60% of Bahawalpur area comprises Cholistan.
- Thar and Nara Desert: It includes Khairpur, Tharparkar, Umer Kot and Sanghar.
- Chagai and Kharan Desert: It is located in Balochistan are extremely dry.

5) COASTAL AREA

Pakistan has a coastline along the Arabian Sea in the southern part of the country, extending approximately 1,046 kilometers (650 miles).

COASTAL AREA OF PAKISTAN

Pakistan has a coastline along the Arabian Sea in the southern part of the country, extending approximately 1,046 kilometers (650 miles). The coastal area of Pakistan encompasses several key features:

1. COASTAL PLAINS

- The coastal plains of Pakistan stretch along the Arabian Sea and are characterized by low-lying terrain with sandy beaches and alluvial deposits.
- These plains support agriculture, particularly the cultivation of crops like rice and sugarcane, through irrigation from the Indus River.

2. INDUS DELTA

- The delta formed by the Indus River as it flows into the Arabian Sea is one of the most prominent features of Pakistan's coastline.
- It is a dynamic and ecologically diverse region, characterized by marshlands, tidal creeks, and mangrove forests.
- The Indus Delta is a crucial habitat for various species of flora and fauna, including migratory birds and marine life.

3. MANGROVE FORESTS

- The coastal areas of Pakistan are home to extensive mangrove forests, particularly in the Indus Delta region.
- Mangroves play a vital role in stabilizing coastal ecosystems, protecting against erosion, and providing habitat for marine species.
- The mangrove forests of Pakistan include species such as *Avicennia marina*, *Rhizophora mucronata*, and *Ceriops tagal*.

4. PORTS AND URBAN CENTERS

- The coastal area is home to several important ports and urban centers, including Karachi, the largest city and commercial hub of Pakistan.
- Karachi Port and Port Qasim are major seaports that handle a significant portion of the country's maritime trade and shipping activities.
- Other coastal cities and towns, such as Gwadar, Ormara, and Pasni, also serve as important centers for fishing, trade, and transportation.

5. MARITIME ACTIVITIES

- The coastal area of Pakistan supports various maritime activities, including fishing, shipping, and naval operations.
- Fishing is an important economic activity, providing livelihoods for coastal communities and contributing to the country's seafood industry.
- The coastline also serves as a strategic location for maritime trade and transportation, connecting Pakistan to global markets.

Overall, the coastal area of Pakistan is not only economically significant but also ecologically diverse, supporting a range of ecosystems and providing valuable resources and habitats for both humans and wildlife.

NORTH EASTER MOUNTAIN REGION OF PAKISTAN

The northeastern mountainous region of Pakistan primarily refers to the area encompassing the northern parts of Khyber Pakhtunkhwa (KP) province and the Gilgit-Baltistan (GB) region. This region is characterized by its stunning mountain landscapes, including parts of the Himalayas, Karakoram Range, and Hindu Kush Range. Here are some key features of this area:

1. HIMALAYAN RANGE

- The northeastern part of Pakistan includes portions of the Himalayan mountain range, which extends into GB and the Chitral District of KP.
- These mountains feature steep slopes, deep valleys, and lush alpine forests.
- Notable peaks in this range include Nanga Parbat, the ninth-highest mountain in the world, and Rakaposhi.

2. KARAKORAM RANGE

- The northeastern region is dominated by the Karakoram Range, renowned for its jagged peaks, glaciers, and high-altitude landscapes.
- Some of the world's highest peaks are located here, including K2, the second highest mountain globally, and Broad Peak.
- The Karakoram Highway (KKH), one of the highest paved international roads, connects Pakistan with China through this mountain range.

3. GILGIT-BALTISTAN (GB) REGION

- GB is a mountainous region in the northeast of Pakistan, bordered by China to the north and KP province to the south.
- It is known for its breathtaking scenery, including snow-capped peaks, turquoise lakes, and picturesque valleys.
- The region is home to several ethnic groups, including the Balti, Shina, and Burusho people, each with their own distinct cultures and languages.

4. GLACIERS AND RIVERS

- The northeastern mountainous region of Pakistan contains numerous glaciers, which are vital sources of freshwater for the region and beyond.
- Glacial meltwater feeds into rivers such as the Indus, Gilgit, and Hunza, providing water for irrigation, hydropower generation, and other uses downstream.

5. TREKKING AND MOUNTAINEERING

- The area is a popular destination for trekking, mountaineering, and adventure tourism.
- Tourists from around the world visit this region to trek to base camps of some of the highest peaks, including K2 and Nanga Parbat, and to explore remote valleys and villages.

6. CULTURAL DIVERSITY

- The northeastern mountainous region is culturally diverse, with various ethnic groups practicing unique traditions and lifestyles.
- Traditional practices such as farming, animal husbandry, and handicrafts are prevalent in many mountain communities.

Overall, the northeastern mountainous region of Pakistan is not only a geological marvel but also a cultural treasure, offering breathtaking natural beauty and rich cultural experiences to visitors and residents alike.

NORTH WESTERN MOUNTAIN REGION OF PAKISTAN

The northwestern and western mountainous region of Pakistan encompasses parts of the Khyber Pakhtunkhwa (KP) province and the Balochistan province. This rugged and mountainous terrain is characterized by its unique landscapes, diverse ethnicities, and strategic significance. Here are some key features of this region:

1. HINDU KUSH RANGE

- The northwestern part of Pakistan includes segments of the Hindu Kush Range, which extends from Afghanistan into KP and northern Balochistan.
- These mountains feature rugged peaks, deep valleys, and high plateaus.
- The Hindu Kush acts as a natural barrier between South Asia and Central Asia.

2. SULAIMAN RANGE

- The Sulaiman Range runs through southwestern KP and eastern Balochistan.
- It consists of rugged mountains and dry valleys, with peaks reaching elevations of over 3,000 meters (9,800 feet).
- The range serves as a geographical and cultural divide between the Indian subcontinent and the Iranian Plateau.

3. ETHNIC DIVERSITY

- o The northwestern and western mountainous region is home to a diverse array of ethnic groups, including Pashtuns, Baloch, and other smaller communities.
- o Each ethnic group has its own distinct language, culture, and traditions, contributing to the region's rich cultural tapestry.

4. STRATEGIC IMPORTANCE

- o The mountainous terrain of this region has significant strategic importance due to its proximity to Afghanistan and Iran.
- o It has been a historically significant crossroads for trade and invasions, and today it plays a crucial role in regional geopolitics.

5. NATURAL RESOURCES

- o The region is rich in natural resources, including minerals such as coal, copper, and chromite.
- o Mining and extraction activities are prevalent in some areas, contributing to the local economy but also raising environmental concerns.

6. CHALLENGES AND OPPORTUNITIES

- o The rugged terrain and limited infrastructure present challenges for development and connectivity in this region.
- o However, the region also offers opportunities for eco-tourism, adventure sports, and cultural exchanges, attracting visitors interested in exploring its scenic beauty and diverse cultures.

Overall, the northwestern and western mountainous region of Pakistan is characterized by its rugged beauty, cultural diversity, and strategic significance. While facing challenges, it holds immense potential for sustainable development and cultural exchange.

GEOGRAPHICAL IMPORTANCE OF PAKISTAN

Pakistan holds significant geographical importance due to its strategic location, diverse landscapes, and natural resources. Here are some key aspects of its geographical importance:

1. STRATEGIC LOCATION

- o Pakistan is situated at the crossroads of South Asia, Central Asia, and the Middle East, making it a bridge between different regions.
- o Its location provides access to important sea routes, including the Arabian Sea and the Indian Ocean, facilitating trade and maritime activities.
- o Pakistan shares borders with several countries, including China, India, Afghanistan, and Iran, giving it geopolitical significance in regional affairs.

2. TRADE AND TRANSPORT HUB

- o Pakistan's location at the crossroads of major trade routes, such as the Silk Road and the Arabian Sea trade route, has historically made it an important center for trade and commerce.
- o The country's ports, including Karachi Port and Gwadar Port, serve as crucial gateways for maritime trade, connecting South Asia with the Middle East, Africa, and beyond.

- Pakistan's road and rail networks provide vital links for overland trade between South Asia, Central Asia, and China.

3. NATURAL RESOURCES

- Pakistan is rich in natural resources, including minerals such as coal, natural gas, copper, and gold.
- Its fertile plains, particularly in the Punjab and Sindh provinces, support agricultural activities, making Pakistan an important agricultural producer in the region.
- The country's diverse landscapes, including mountains, deserts, and coastal areas, offer opportunities for tourism, mineral extraction, and energy production.

4. WATER RESOURCES

- The Indus River and its tributaries flow through Pakistan, providing water for irrigation, hydroelectric power generation, and domestic use.
- The Indus Basin Irrigation System, one of the largest in the world, supports agriculture in the country's fertile plains and contributes to food security.

5. SECURITY AND STABILITY

- Given its location at the nexus of regional conflicts and geopolitical rivalries, Pakistan's stability is crucial for regional security.
- The country's role in counterterrorism efforts and its relationship with neighboring countries, particularly Afghanistan and India, have implications for peace and stability in the region.

Overall, Pakistan's geographical importance stems from its strategic location, natural resources, and role in regional trade and security dynamics. Understanding and harnessing these geographical advantages are essential for the country's economic development, regional cooperation, and geopolitical influence.